AfriPunctCap: Pan-African Speech-Pause Alignment and Fast CPU Punctuation Restoration Across 314 African Languages

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***Abstract*—*In African language orthographies, writers conventionally omit commas at natural clause, list, and conjunction boundaries, despite speakers making distinct, audible breath pauses (350–600 ms) in natural speech. This discrepancy creates a severe bottleneck for Text-to-Speech (TTS) models, which rely on punctuation to produce natural phrasing, and for Automatic Speech Recognition (ASR) outputs, which lack punctuation altogether. In this work, we introduce **AfriPunctCap**, an end-to-end framework for acoustic pause detection and punctuation/capitalization restoration spanning **314 African languages**. Using Meta's Massively Multilingual Speech (MMS) forced aligner, we align 292,791 audio clips (950+ hours) from the Afri2Universal corpus to detect true acoustic pauses (≥ 350 ms) and insert 400,000+ ground-truth commas. We structure these transcripts into multi-sentence packed streams to train afro-xlmr-base with multi-task dual heads for simultaneous capitalization and punctuation. Evaluated across 20 core African languages spanning Niger-Congo, Afroasiatic, and Nilo-Saharan families, our model achieves **0.72–0.92 Comma F1** and **0.86–0.99 Period F1**. Finally, we show that non-autoregressive token classification vastly outperforms sequence-to-sequence models (e.g., T5) by preventing comma under-generation, and deploy an INT8 quantized ONNX engine running in **15–25 ms on CPU** with an open-source serverless demonstration interface.*

1. Introduction

Over two thousand languages are spoken across the African continent, representing immense linguistic diversity. However, modern speech technologies—specifically Text-to-Speech (TTS) synthesis and Automatic Speech Recognition (ASR)—face a fundamental mismatch: the discrepancy between *conventional written orthography* and *natural spoken prosody*.

In languages such as Akan (Twi), Yoruba, Swahili, Hausa, and Amharic, writers habitually omit commas before coordinating conjunctions (e.g., *ne* in Akan, *àti* in Yoruba, *na* in Swahili), complementizers (s■), appositives, and compound titles. In contrast, native speakers consistently insert acoustic pauses ranging from 350 ms to over 600 ms at these syntactic junctions. Without explicit commas, neural TTS engines synthesize speech without pausing, producing rushed and unnatural audio. Simultaneously, downstream ASR applications receive unpunctuated text streams that degrade machine translation and entity extraction.

To solve this problem, we present **AfriPunctCap**: (1) we acoustically align 292,791 speech recordings across 314 African languages to detect speech pauses (≥ 350 ms) and insert 400,000+ ground-truth commas; (2) we curate multi-sentence sequences (1–3 sentences packed) to teach models internal sentence-boundary detection; (3) we fine-tune Davlan/afro-xlmr-base with dual multi-task heads; and (4) we deploy an INT8 quantized ONNX engine running in 15–25 ms on CPU.

2. Methodology

2.1 Large-Scale Acoustic Pause Alignment

We leverage speech recordings from the Afri2Universal corpus, covering 314 languages and 950+ hours of audio. For each audio-transcript pair, forced alignment is performed via Meta's MMS CTC Aligner with universal Romanization (uroman). For each adjacent word pair (w_i, w_{i+1}), the silent gap is measured as $\Delta t_i = t_{i+1}^{start} - t_i^{end}$. When $\Delta t_i \geq 350$ ms, a speech pause is identified and resolved into a comma. Across 314 languages, 457,704 speech-pause commas were detected.

2.2 Multi-Sentence Sequence Packing

To prevent *boundary bias* (where models only learn to predict period tokens on the final token of isolated sentences), we construct multi-sentence sequences by concatenating 1, 2, or 3 consecutive sentences from the same language. All punctuation is stripped and casing is lowercased in the input, while targets retain internal sentence-ending periods, commas, and capitalization. From 292,533 sentences, we construct 145,812 multi-sentence training sequences.

2.3 Multi-Task Sequence Tagging Architecture

We adopt Davlan/afro-xlmr-base (278M parameters, 12 layers, 768 hidden dimension) with dual classification heads: (1) Capitalization Head: 3 classes (LOWER, UPPER, ALL_CAPS); (2) Punctuation Head: 5 classes (O, COMMA, PERIOD, QUESTION, EXCLAMATION). Loss is computed on the first subword token of each word with class weights: w_cap = [1.0, 1.6, 2.0] and w_punct = [0.6, 2.2, 1.8, 2.5, 2.5].

3. Experimental Evaluation & Results

Training was executed on a single NVIDIA H200 GPU (140 GB VRAM) using bfloat16 mixed precision for 50 epochs (102,550 steps) at ~1,450 seq/s (~105 minutes total). The table below details per-language performance across 20 representative languages.

Language	Family	Prec.	Rec.	Comma F1	Period F1
Swahili (swh)	Niger-Congo (Bantu)	0.885	0.932	**0.908**	**0.988**
Amharic (amh)	Afroasiatic (Ethiopic)	0.905	0.930	**0.917**	**0.993**
Zulu (zul)	Niger-Congo (Bantu)	0.845	0.966	**0.902**	**0.986**
Hausa (hau)	Afroasiatic (Chadic)	0.809	0.927	**0.864**	**0.959**
Yoruba (yor)	Niger-Congo (Defoid)	0.724	0.855	**0.784**	**0.861**
Lingala (lin)	Niger-Congo (Bantu)	0.779	0.928	**0.847**	**0.840**
Xhosa (xho)	Niger-Congo (Bantu)	0.750	0.874	**0.807**	**0.980**
Shona (sna)	Niger-Congo (Bantu)	0.585	0.859	**0.696**	**0.912**
Somali (som)	Afroasiatic (Cushitic)	0.811	0.914	**0.860**	**0.963**
Tigrinya (tir)	Afroasiatic (Ethiopic)	0.748	0.917	**0.824**	**0.948**
Akan / Twi (aka)	Niger-Congo (Kwa)	0.635	0.833	**0.721**	**0.940**
Ewe (ewe)	Niger-Congo (Gbe)	0.720	0.822	**0.767**	**0.918**
Kinyarwanda (kin)	Niger-Congo (Bantu)	0.793	0.938	**0.859**	**0.954**
Luganda (lug)	Niger-Congo (Bantu)	0.695	0.860	**0.769**	**0.915**
Wolof (wol)	Niger-Congo (Senegambian)	0.686	0.817	**0.746**	**0.842**
Dyula (dyu)	Niger-Congo (Mande)	0.786	0.872	**0.826**	**0.918**
Mooré (mos)	Niger-Congo (Gur)	0.668	0.903	**0.768**	**0.910**
Bemba (bem)	Niger-Congo (Bantu)	0.649	0.858	**0.739**	**0.933**
Dangme (ada)	Niger-Congo (Kwa)	0.631	0.854	**0.725**	**0.909**
Acholi (ach)	Nilo-Saharan	0.558	0.805	**0.659**	**0.823**
Core Languages Avg.	**Pan-African Benchmark**	**0.733**	**0.883**	**0.799**	**0.934**

4. Ablation: Sequence Tagging vs. Autoregressive Seq2Seq

We compared sequence tagging against an autoregressive seq2seq model (T5-Small, 60M parameters) using reversible ASCII proxy representations. T5-Small suffered from severe comma under-generation (0.061 Comma F1) because standard cross-entropy loss rewards verbatim word copying over sparse punctuation. In contrast, sequence tagging with explicit class weighting guarantees 0% hallucination risk and achieves 0.799 Comma F1 and 0.934 Period F1.

5. Edge Deployment & Quantization Analysis

To enable accessible deployment across Africa where high-end GPU infrastructure is scarce, we perform dynamic INT8 quantization on the exported ONNX model. The table below compares the full GPU precision model (bfloat16) against the quantized CPU engine (INT8).

Precision	Comma F1	Period F1	Memory Footprint	Inference Latency
bfloat16 / FP16 (GPU)	**0.799**	**0.934**	540 MB	~3.2 ms (H200 GPU)
INT8 Quantized (CPU ONNX)	0.731	0.915	**278 MB (2x smaller)**	**15–25 ms (Commodity CPU)**

As shown, INT8 dynamic quantization retains **over 91% of comma accuracy** and **98% of sentence-boundary accuracy** while cutting the model size in half and executing in 15–25 ms on standard 2-core x86 CPUs. We deploy the INT8 engine on Modal with a 60-second scale-to-zero window, connected to a free static HTML/JS demonstration interface on Hugging Face Spaces.

6. Conclusion

AfriPunctCap bridges the critical gap between written conventions and spoken pauses across 314 African languages. By combining large-scale acoustic pause alignment with multi-sentence sequence tagging, the model unlocks natural breath pauses for TTS engines and real-time punctuation for ASR transcripts on commodity CPUs. All datasets, models, and code are open-sourced under AfriSpeech.